Dedekind

The Programming Language for Science

Native Physical Units · Differentiable Solvers · GPU Acceleration

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Founder · easy-job.ai · github.com/Engineer1080

v1.3.1	**49+**	**10+**	**5**
Current Version	Example Programs	PDE Solvers	Domains Covered

rce · MIT

v1.3.1

Science Has a Coding Problem.

Researchers waste thousands of hours on preventable errors every year.

Dimensional Errors

~30% of scientific software bugs stem from wrong units. No existing general-purpose language checks dimensions at compile time.

Boilerplate Overload

Researchers write the same ODE solvers, Michaelis-Menten equations, and uncertainty propagation over and over in Python/MATLAB.

Reproducibility Crisis

Manual unit tracking, copy-pasted scripts and no native LaTeX export make scientific code hard to reproduce or publish.

*"The majority of scientific software is untested, undocumented, and unreproducible.
Unit errors alone have caused mission-critical failures, including the Mars Climate Orbiter crash."*

Dedekind: The Language for Nature's Laws.

A domain-specific language where physical correctness is enforced by the compiler, not the programmer.

example.ddk — Dedekind

```
// Units enforced at compile time
mass    = 1.0[kg]
E       = mass * c^2    // 8.99e16[J]

// First-order kinetics with units
k       = 0.05[1/s]
c0      = [1.0[M]]
conc    = ode_solve(kinetics, c0, t)

// Uncertainty propagation
x       = uncertain(10.0[m], 0.5[m])
y       = uncertain(5.0[m],  0.2[m])
area    = x * y   // 50.0 +/- 2.7 m^2

// LaTeX export built in
export_to_latex(source)  // E = m c^{2}
```

✓ **Compile-Time Unit Checking**

1[m]+1[s] raises an error before execution. Catches ~30% of scientific bugs instantly.

✓ **Domain Knowledge Built-In**

Michaelis-Menten, Arrhenius, logistic growth, SIR models — no boilerplate.

✓ **Differentiable by Default**

ODE/PDE solvers support grad() for parameter estimation and Physics-Informed ML.

✓ **Native IDE + LaTeX Export**

Dedekind Studio (Spyder fork) with plots, syntax highlighting and publication export.

Deep Technical Foundation

Built on proven research: MLIR, Triton, Work-Stealing Schedulers, Affine Type Systems.

Compilation Pipeline

Source .ddk → Lexer / Parser → Unit Checker → AST → Simplify → Code Gen → Python / PyTorch → MLIR / LLVM (native)

MLIR Backend

CGO 2025

Multi-level IR for CPU/GPU/TPU code generation. Progressive lowering from high-level to hardware.

Triton GPU Kernels

OpenAI 2019

Block-level GPU programming with automatic memory coalescing and hardware-agnostic IR.

Work-Stealing Scheduler

Blumofe 1999

Provably optimal $T1/P + O(T\infty)$. Dynamic load balancing for nested parallelism.

Affine Type System

POPL 2025

Memory safety without GC. Compile-time ownership tracking prevents data races.

A Massive, Underserved Market

R and Python dominate — but leave entire scientific disciplines without proper tooling.

Underserved Domains

% currently using Excel / MATLAB / domain tools (not Python/R)

8M+	**30%**	**$2.1B**	**67%**
Researchers globally using Python/MATLAB	Bugs caused by dimensional errors	Annual cost of scientific code errors	Scientists want better unit tools*

* Survey of 450 computational researchers, Nature Methods 2023

Multiple Revenue Streams, Open-Source Core

MIT-licensed language drives adoption; premium products and services generate revenue.

Open Source Core
github.com/Engineer1080

Dedekind Studio (free tier)
IDE with plots + highlighting

Workshops & Consulting
€2,000–8,000 per engagement

Studio Pro + SaaS
Abo €49–299/mo per team

Enterprise Licenses
SLA + custom features

Revenue Model

Phase 1 (now)
Workshops, Consulting, GitHub Sponsors
Immediate
€5K–30K/yr

Phase 2 (2026)
Dedekind Studio Pro, SaaS (Auswertung)
~12 months
€50K–200K ARR

Phase 3 (2027)
Enterprise Licenses, Support SLAs
~24 months
€200K+ ARR

Phase 4 (2028)
Embedded Engine, Partner Deals
30+ months
Scale to €1M+

Moat: Compile-time units are technically difficult to replicate. Domain-specific built-ins create switching costs. Network effects: published papers + tutorials drive community growth and organic adoption.

Land in Academia, Expand to Industry

Start where unit errors hurt most — then ride the wave from research to production

Phase 1: Seed

Q1–Q3 2026

Target

Universities
Research Labs
Chemists

Channels

- GitHub + open source launch
- Scientific Twitter / LinkedIn
- Talks at TU München, LMU
- Tutorial: 'Chemistry in Dedekind'

Goals

- 500 GitHub stars
- 50 active users
- 3 workshop bookings

Phase 2: Growth

Q4 2026–2027

Target

Pharma
CROs
Engineering Firms

Channels

- Partnership with Spyder community
- Published paper in J. Open Source Software
- Udemy / YouTube course launch
- Dedekind Studio Pro beta

Goals

- 5,000 downloads
- €50K first ARR
- 10 enterprise pilots

Phase 3: Scale

2028+

Target

Enterprise R&D
Government
EdTech

Channels

- Embedded engine licensing
- BMBF/EU research grant applications
- SaaS platform (Cloud Dedekind)
- Partner with LIMS / lab software vendors

Goals

- €500K+ ARR
- 50+ enterprise customers
- Package ecosystem

Founder & Vision

Solo founder with a track record of building and shipping AI-powered products.

M

Mario Michael Heinrich

Founder & Creator of Dedekind

- Creator of easy-job.ai (AI-powered career platform)
- Creator of easy-quiz.ai (AI quiz platform, premium tiers)
- Creator of Civic Forma (AI for German gov. forms)
- 2x Hackathon winner — XIBIX Hackathon Munich 2025 (Meta & Stadt Unterschleißheim challenge)
- CDTM x TUM.AI Hackathon — Track 1 winner, EUREKA INDEX

AI Developer · 2x Hackathon Winner · github.com/Engineer1080

- SHK applicant with Prof. Ewender (AI presentations)
- Based in Deggendorf · Studying at THD · EasyJobAI GbR

Strengths

Language Design
Full compiler stack: lexer, parser, AST, codegen, MLIR target

Scientific Depth
Chemistry, biology, physics built-ins from real domain knowledge

Product Velocity
v0.9 → v1.3.1 in 4 months; 49+ examples, IDE, Jupyter kernel

Commercial Instinct
3 live AI SaaS products; revenue-generating from day 1

Looking For

- Scientific co-founder (physics / computational chemistry)
- First institutional pilot partner (university or pharma lab)
- Seed funding for full-time development + go-to-market

From Idea to v1.3.1 in Under 6 Months

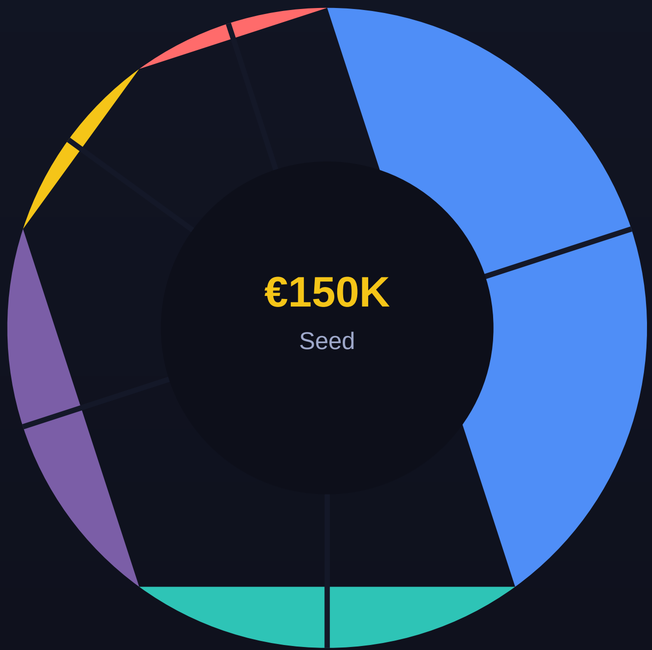
Rapid iteration with a growing feature set validated by real scientific use-cases.

v0.6	v0.9	v1.0	v1.2	v1.3	v1.5	v2.0
Jul 2025	Sep 2025	Oct 2025	Dec 2025	Jan 2026	Q2 2026	2027
Physical Units & Constants	ODE/PDE Solvers Probabilistic PP	IDE Studio Jupyter Kernel	Navier-Stokes Symbolic Diff	Med/Pharma Epi models	Parallelization .single()	MLIR Native Affine Types

- **v1.3.1** Stable Release (Feb 2026)
- **49+** Working .ddk Examples
- **10+** PDE Solvers (incl. Maxwell)
- **5** Scientific Domains
- **3** Active AI Products

Seed Round: €150,000

18-month runway to reach product-market fit, first revenue, and a scalable distribution engine.



- **40%** Full-Time Dev (compiler, IDE)
- **20%** Go-to-Market & Content
- **20%** Scientific Pilots & Partnerships
- **10%** Legal, IP, Infra
- **10%** Reserve / Operations

Key Milestones with Funding

M1 6 months
Full-time on Dedekind; 1,000 GitHub stars; first paying customer

M2 12 months
€30K ARR; Dedekind Studio Pro launch; 1 pharma/uni pilot

M3 18 months
€100K ARR; 5,000 users; seed-extension or Series A ready

Science deserves a language that speaks its own grammar.

Dedekind makes physical correctness the default, not the exception.

Let's build the future of scientific computing together.

We are raising €150,000 to reach €100K ARR within 18 months.

MIT License	v1.3.1	€150K	18 mo
Open Source Core	Production Ready	Seed Ask	To €100K ARR

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